



CREDENCE IT PROFESSIONAL TRAINING INSTITUTE, PUNE

Credence - AI-Driven Full Stack Software Testing Master Class/ Software Testing Syllabus

Table of Contents

Table of Contents	0
1. Credence Software Testing Syllabus Process	2
2. Introduction to Software	3
3. Software Development Life Cycle (SDLC)	3
4. Software Testing Fundamentals	4
5. Testing Levels	4
6. Testing Terminology	5
7. Test Documentation Concepts	6
8. Project Management Tool (HP ALM & Jira)	6
9. SQL & Database Concepts	7
10. Advance SQL	8
11. Database types in Syllabus	9
12. Database Testing	9

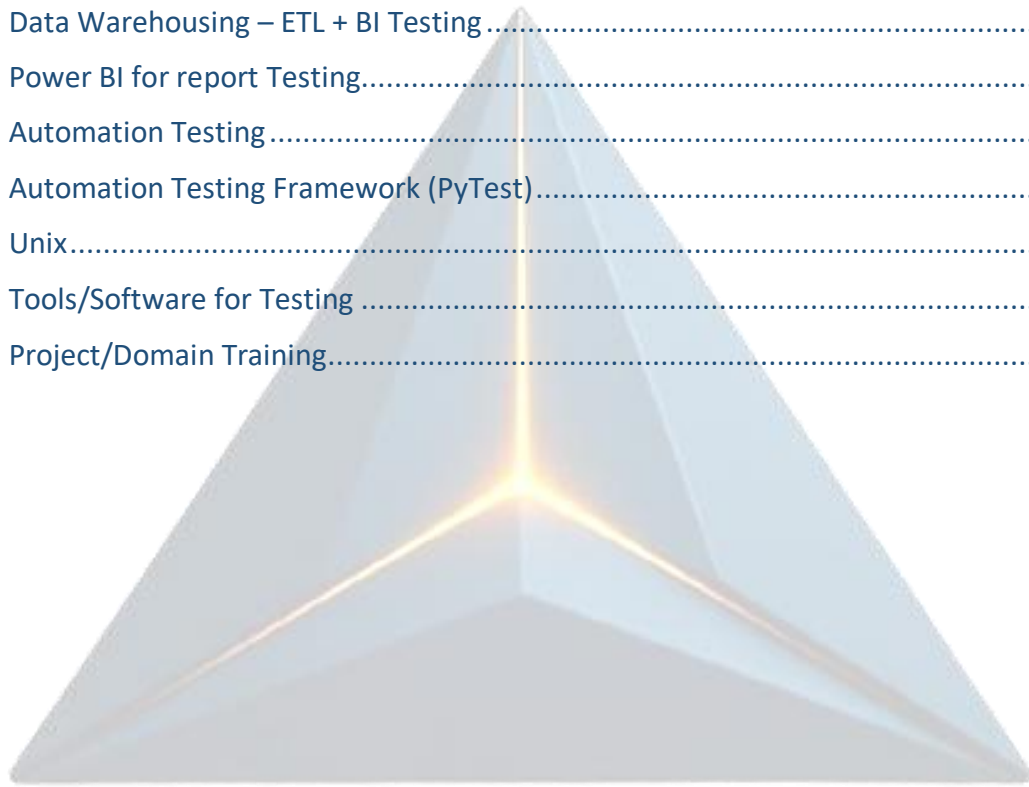
Email: Info@credence.in **Website:** <https://www.credence.in>

Contacts: +91 8237916162, +91 7391053250, +91 9579064658, +91 9091929355



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13.	Azure Cloud Concepts.....	10
14.	Perform testing using JMeter	10
15.	API testing using Postman	11
16.	Salesforce CRM.....	11
17.	Data Warehousing – ETL + BI Testing	12
18.	Power BI for report Testing.....	14
19.	Automation Testing	14
20.	Automation Testing Framework (PyTest).....	18
21.	Unix.....	19
22.	Tools/Software for Testing	20
23.	Project/Domain Training.....	21



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1. Credence Software Testing Syllabus Process



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2. Introduction to Software

- ▲ **What is Software?**
 - ▲ **Types of Software**
 - System Software
 - Application Software
 - ▲ **Bidding in the Software Industry**
 - ▲ **Organization or Team Structure in the Software Industry**
-

3. Software Development Life Cycle (SDLC)

- ▲ **Introduction to SDLC**
- ▲ **Core Phases of SDLC**
 - Planning Phase
 - Requirement Gathering Phase
 - Design Phase
 - Development Phase
 - Testing Phase
 - Deployment Phase
 - Maintenance Phase
- ▲ **Models in SDLC:**
 - Waterfall Model
 - V Model
 - Spiral Model
 - Agile

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4. Software Testing Fundamentals

- ▲ Introduction to Software Testing
 - ▲ Why Testing is Necessary and Important?
 - ▲ Why Developers Can't Do the Testing? Why Do We Need Testers?
 - ▲ Advantages and Disadvantages of Software Testing
 - ▲ What is Error, Defect or Bug, Failure?
 - Reasons and Examples
 - ▲ Comparison Table: Defect, Error, and Failure in a Banking Application
-

5. Testing Levels

- ▲ Unit Testing
 - Basis Path Testing
 - Control Structure Testing
 - Conditional Coverage
 - Loops Coverage
- ▲ Integration Testing Types
 - Incremental Integration Testing
 - Top-Down Integration Testing
 - Bottom-Up Integration Testing
 - Sandwich (Hybrid) Integration Testing
 - Big Bang Integration Testing or Non-Incremental Integration Testing

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▲ System Testing

- Usability
 - GUI Testing
 - Manual Support Testing
- **Functional Testing**
 - Behavioral coverage
 - Input Domain coverage
 - Error handling coverage
 - Backend Coverage
 - Service Level Coverage
 - Calculation based Coverage
- **Non-Functional Testing**
 - Recovering Testing
 - Compatibility Testing
 - Configuration Testing or Hardware Compatibility Testing
 - Inter-System Testing
 - Installation Testing
 - Parallel Testing
 - Globalization Testing
 - Sanitation/Garbage Testing
 - Performance Testing
 - Security Testing

▲ User Acceptance Testing (UAT)

- Alpha Testing
- Beta Testing

6. Testing Terminology

- ▲ Static Testing vs Dynamic Testing
- ▲ Quality Assurance (QA) vs Quality Control (QC)

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- ▲ White Box Testing vs Black Box Testing vs Grey Box Testing
- ▲ Software Testing Life Cycle (STLC) -Sign off email
- ▲ Environments
- ▲ Defect Life Cycle
- ▲ Test Execution Life Cycle
- ▲ Smoke testing and Sanity Testing
- ▲ Re-Testing and Regression Testing
- ▲ Exploratory Testing
- ▲ Ad-hoc Testing
- ▲ Monkey Testing
- ▲ Positive Testing
- ▲ Negative Testing

7. Test Documentation Concepts

- ▲ Test Plan
- ▲ Test Scenarios
- ▲ Test Case

8. Project Management Tool (HP ALM & Jira)

- ▲ Writing Test Cases
- ▲ Release/Cycle Management

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- ▲ Executing Test Cases
- ▲ Raising Defects
- ▲ Severity and Priority
- ▲ Traceability Matrix
- ▲ Defect Management

9. SQL & Database Concepts

- ▲ What is Data?
- ▲ What is a Database?
- ▲ DBMS vs RDBMS
- ▲ SQL Overview
 - SQL Syntax and Tables
 - SQL Data Types
 - Where Clause in SQL
 - Basic SQL Queries
- ▲ Installation
 - Oracle Database
 - SQL Developer
 - Oracle DB Online Compiler
- ▲ SQL Commands and Functions
- ▲ DDL, DML, DCL, TCL Commands

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- Insert, Update, Delete, Select
- ▲ **SQL Operators**
 - ▲ Arithmetic, Comparison, Logical Operators, EXISTS and NOT EXISTS
- ▲ **SQL Constraints**
 - Table-Level and Column-Level Constraints (Primary Key, Foreign Key, etc.)
- ▲ **SQL Functions**
 - Aggregate
- ▲ **SQL Joins**
 - Inner Join, Left Outer Join, Right Outer Join, Full Outer Join
- ▲ **SQL Clauses**
 - Where, Group By, Having, Order By
- ▲ **Pseudo Columns**
 - Rank(), Dense_Rank(), Rownum, Rowid
- ▲ **Subquery and Aggregation Functions**
- ▲ **Normalization**
- ▲ **Case statement**
- ▲ **Finding and Deleting Duplicate Data**

10. Advance SQL

- ▲ Store procedure
- ▲ Trigger
- ▲ Views
- ▲ Materialized views

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- ▲ Partition by and group by
- ▲ With clause
- ▲ CTE
- ▲ Coalesce function
- ▲ NVL
- ▲ Trim / Ltrim / Rtrim
- ▲ Round
- ▲ Cast
- ▲ Pivoting
- ▲ Query optimization concept
- ▲ Character, Date, Conversion Functions
- ▲ Temporary tables
- ▲ Replace
- ▲ Regression expressions
- ▲ ROW_NUMBER(), RANK(), DENSE_RANK(), LAG(), and LEAD()
- ▲ Indexing

11. Database types in Syllabus

- ▲ Oracle
- ▲ Microsoft SQL server
- ▲ Azure-Cloud database

12. Database Testing

- ▲ What is Database Testing?
- ▲ Why Do We Do Database Testing?
- ▲ Benefits of Database Testing
- ▲ Database Testing List

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13. Azure Cloud Concepts

- ▲ What is Cloud Computing?
- ▲ Types of Cloud (PaaS, SaaS, IaaS)
- ▲ Introduction to Microsoft Azure
- ▲ Azure Data Bricks
 - Notebooks, SQL, Cluster, Query Run, Data Loading, Permissions
- ▲ Azure Data Factory
 - Pipeline Creation and Validation, Server Checking, Monitoring
- ▲ Azure Synapse
 - SQL Query, Data Frames, Query Validation
- ▲ Azure DevOps
 - Sprint Creation, Story Creation, Bug Tracking, Link with Story

14. Perform testing using JMeter

- ▲ What is Performance Testing
- ▲ Types of Performance Testing
 - 1. Stress Testing
 - 2. Load Testing
- ▲ Overview of JMeter (Performance Testing Tool)
- ▲ Test Plan, Threads, Assertions, Listeners, Https request, Http request.
- ▲ Overview of Blaze meter (Automation tool used for Performance testing)
 - 1. How to Generate JMeter script using Blaze meter
 - 2. Automation Test Scenarios

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- ▲ Performance Testing on Non-GUI Mode.
- ▲ Performance Dashboard Creation using non-GUI Mode

15. API testing using Postman

- ▲ Overview of Postmen tool (API Testing tool)
- ▲ Types of Request
 - 1. GET Request
 - 2. POST Request
 - 3. PUT Request
 - 4. DELETE Request
- ▲ Login Functionality Testing using Postman
 - 1. No Auth
 - 2. Basic Auth
 - 3. Bearer Token
- ▲ Integration Testing using Postman Tool

16. Salesforce CRM

- ▲ What is Salesforce ?
- ▲ What is Cloud ?
- ▲ Types of Cloud in Salesforce CRM
- ▲ Sales Cloud, Service Cloud, Commerce Cloud, Analytical
- ▲ Cloud, Experience Cloud, Marketing Cloud
- ▲ Dierent Terminology in Salesforce
- ▲ What is CRM?
- ▲ Roles and Responsibility of Salesforce Admin
- ▲ Salesforce Reports and Dashboard
- ▲ Lead, Account, Contact & Opportunities
- ▲ UI Testing in Salesforce
- ▲ Trigger Testing in Salesforce7

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17. Data Warehousing – ETL + BI Testing

▲ Introduction to Data Warehousing & ETL + BI Testing

- Transactional Vs Analytical System
- Introduction Data Warehouse
- What is Data warehouse?
- What is ETL?
- Data Warehouse Architecture/technical Flow/Data Flow/High level Design (HLD)
- What is Data mart?
- Differences between OLTP and OLAP.
- Table Structure validation/ Metadata Testing and its email communication
- OLTP vs OLAP
- Normalization and De-Normalization
- Relational vs Dimensional Modeling
- Fact Table and Dimension Table
- Types of Facts and Dimensions
 - Additive, Semi-Additive, Non-Additive Facts
 - SCDs (Slowly Changing Dimensions) Types 0, 1, 2, 3, 4, 6
 - Junk, Conformed, Degenerated Dimensions
- Surrogate Key
- Star Schema, Snowflake Schema, Galaxy Schema

▲ ETL Tools and SQL Server/Oracle IDE

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- SQL Server Integration Services (SSIS)
- Informatica
- SQL Server Management Studio (SSMS)

▲ ETL and Report Testing

- What is ETL Testing and importance of ETL testing
- Difference between ETL Testing and Database Testing
- ETL testing scenarios and Test cases
- Data Mapping Documents/God's Documents/ Data Leakage Documents.
- Story of EMP_ETL Project.
- EMP_ETL Project Test Scenarios
- Queries to Compare Data
- Live data/Test Data
- Loading type, Initial, incremental, Full Load
- ETL Testing Challenges
- Report Testing and Tableau Introduction
- Report Testing Challenges

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▲ Types of ETL Testing

- Metadata Testing
- Data Completeness Testing

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- Data Transformation Testing
 - Data Quality Testing
-

18. Power BI for report Testing

- ▲ Introduction
- ▲ Need to Power BI
- ▲ Testing Aspects
- ▲ Scenario for functional and integration testing
- ▲ Report testing
- ▲ KPI testing
- ▲ Question and Answer

19. Automation Testing

- ▲ Introduction to Automation Testing
- ▲ Python Basic Syntax
- ▲ Data Types
 - Numbers
 - Strings
 - Booleans
 - Lists
 - Tuples
 - Sets
 - Dictionaries

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▲ Variables

- Defining variables
- Variable naming rules
- Constants

▲ Operators

- Arithmetic operators
- Comparison operators
- Logical operators
- Assignment operators
- Membership and identity operators

▲ Control Flow

- Conditional Statements
 - if, elif, else
- Loops
 - for loop
 - while loop
 - Loop control (break, continue, pass)

▲ Functions

- Defining Functions
 - - Syntax and calling functions
 - - Parameters and arguments

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- - Return values
- Lambda Functions
- Built-in Functions
- Modules and Packages
 - - Importing modules
 - - Creating and using packages
- ▲ **File Handling**
 - Reading and Writing Files
 - open (), read (), write () methods
 - Working with text and binary files
 - File Methods
- ▲ **Exception Handling**
- ▲ **Object-Oriented Programming (OOP)**
 - Class, Object, Constructor, Methods, Inheritance, Encapsulation, Polymorphism
- ▲ **Selenium Components**
- ▲ **Installation of Selenium, Python IDE, and Web Driver Setup**
- ▲ **Selenium WebDriver**
 - **Browser Invocation**
 - Chrome

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- Firefox
- Edge
- **Locators**
 - ID
 - Name
 - Class Name
 - Tag Name
 - Link Text
 - Partial Link Text
 - CSS Selector
 - XPath
- **Working With Elements**
 - Alert
 - Calendar and Time Picker
 - Checkbox
 - Click and Hold
 - Double Click
 - Drag and Drop
 - Dropdown
 - Element Displayed
 - Explicit Wait
 - File Upload
 - iFrame
 - Implicit Wait
 - Mouse Hover
 - Radio Button
 - Right Click (Context Click)
 - Switch Tab

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- Switch Window
- Web Table
- Web Table Fixed Header

20. Automation Testing Framework (PyTest)

- ▲ **What is a Framework?**
- ▲ What is Pytest Framework?
- ▲ **Pytest Framework**
 - Parallel Run, Markers, HTML Reports
 - Integration with Jenkins and GitHub
 - Test Case Writing, Conftest File
 - Test Data Using Params and Excel-DDT
 - Log Generation
 - Allure Reports
- ▲ Pytest Project - parallel run, markers, html reports
- ▲ Pytest Project integration with Jenkins
- ▲ Pytest Project integration with GitHub
- ▲ Project- standalone code
- ▲ Project - framework implementation from scratch and folder creation
- ▲ Project - page objects
- ▲ Project - test case
- ▲ Project - conftest file

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- ▲ Project - config file to replace hard values
- ▲ Project - multiple browser
- ▲ Project - html reports
- ▲ Project - parallel run and markers
- ▲ Project - log generation
- ▲ Project - test data using params
- ▲ Project - test data using excel-ddt
- ▲ Project - allure reports
- ▲ Project - Jenkins
- ▲ Project - GitHub

21. Unix

- ▲ Unix and Linux deference
- ▲ Basic commands in Unix
- ▲ Unix file structure/architecture
- ▲ Create commands for file
- ▲ Open commands for file
- ▲ Copy commands for file
- ▲ Head and tail commands
- ▲ Grep and grep commands
- ▲ SED commands

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- ▲ CH mod
- ▲ Find or locate files

22. Tools/Software for Testing

1. SQL Developer
2. SQL server management studio
3. Visual studio +SSIS
4. Tableau
5. Power-BI
6. HPALM
7. Unix-Putty
8. PyCharm
9. Jira
10. Git
11. Jenkins
12. Postman
13. JMeter
14. Salesforce CRM
15. Azure Databricks
16. Notepad ++
17. Excel

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18. Outlook

19. Chat-GPT

20. Deepseek

21. Grok-AI

23. Project/Domain Training

- ▲ **Telecom Billing/CRM/OSS/Mediation**
- ▲ **Capital Market/Investment Banking**
- ▲ **Healthcare Domain**
- ▲ **Card Payment Systems**
- ▲ **Life Sciences**
- ▲ **Insurance Domain**
- ▲ **Automotive Domain**

Note: 3-4 domain for each batch

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